Kepler’s laws

antocciphysics | TEACHER KEY | CPA

Learning target: Use orbital shape, equal areas and period comparisons.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 11; 3 course days later.*
3 s to top; 6 s total, assuming no drag and equal launch/landing heights.

02. Explain a worked example (worked)
Faster near perihelion, slower near aphelion; Sun at focus.

03. Look and explain (model)
Fastest closest; slowest farthest. Sun at focus, not center.

04. Build the solution (practice)
Outer; 16 years.

05. Use the idea (reason)
No; need orbital phase, time reference, geometry, and Earth’s position, with uncertainty.

06. Choose a method (mixed) *Review: Unit 2, Day 2; 12 course days later.*
50 m/s downward.

07. Exit check (exit)
A; equal areas in equal times.

08. STRETCH: Apply it another way (stretch)
Period grows faster than a: $T^2=a^3$, so the given size factor 4 produces period factor 8.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 11; 3 course days later.*
Both ranges≈68.94 m. Times≈2.536 s and 5.438 s. The larger angle gives longer flight.

02. Explain a worked example (worked)
Faster near perihelion, slower near aphelion; Sun at focus.

03. Represent and interpret (model)
Fastest closest; slowest farthest. Sun at focus, not center.

04. Solve and justify (practice)
$T=\sqrt{1.42^3}\approx1.69$ years; 8 years at 4 AU; $T\propto a^{1.5}$.

05. Analyze the reasoning (reason)
No; need orbital phase, time reference, geometry, and Earth’s position, with uncertainty.

06. Choose a method (mixed) *Review: Unit 2, Day 2; 12 course days later.*
$v_f=34$ m/s down; $\Delta y=57$ m down.

07. Exit check (exit)
$2^{1.5}\approx2.83$; central mass/unit normalization differs.

08. STRETCH: Apply it another way (stretch)
a=9 AU; perihelion 4.5 AU, aphelion 13.5 AU. Formula is supplied for the extension.